

Lab 38 Worksheet — Password Strength Analysis and Authentication Policy

Name: ____ Date: ____

Exam objective: Analyse password strength quantitatively and write an authentication policy grounded in entropy and crack-time evidence (Core 2 objectives 2.2 and 2.6).

Record as you go

Step	What you did	What you observed
1	Open https://alfredang.github.io/cybersecuritysimulator/ and select Password Lab from the top menu.	
2	Enter the password 'password' and record the strength indicator, the estimated crack time and the approximate entropy in bits.	
3	Enter 'Password1' and record the same three metrics, noting how little a capital letter and a digit actually add.	
4	Enter 'P@ssw0rd!' and record the metrics, then note that character substitution on a dictionary word is exactly what cracking tools try first.	
5	Enter a 16-character passphrase such as 'correct horse battery staple' and record how the metrics change against the substituted password.	
6	Enter a 20-character random string and record the metrics, then compare the entropy against the passphrase to see which strategy wins.	
7	Build the analysis table with all five passwords, their length, character set size, entropy in bits and estimated crack time.	
8	State the conclusion the evidence supports:	

Step	What you did	What you observed
	length increases entropy far more effectively than character substitution, so a long passphrase beats a short complex password.	
9	Review the weak and strong password examples the tool lists, and record the pattern that makes the weak ones weak.	
10	Write the authentication policy: minimum length, complexity requirements, expiry rules, reuse prevention, and the account lockout threshold and duration.	
11	Justify each policy rule with a measured figure from your table rather than a general assertion.	
12	Add the MFA requirement to the policy, stating which factor categories are acceptable and which accounts must use MFA without exception.	

Verification

Success criterion: Your analysis table records the actual entropy and crack time from the tool for all five passwords; the conclusion that length beats substitution is supported by your own figures; and every policy rule cites a measured value.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
